

L-21: Generating Functions

Non-negative Integer-valued Random Variables

Discrete RVs with non-negative integer values.

Common discrete distributions:

- Bernoulli
- Binomial
- Geometric
- Hypergeometric
- Negative Binomial
- Poisson
- Discrete Uniform

For these, $\sum_{n=0}^{\infty} p_n = 1$.

Probability Generating Functions (PGF)

Let X be a random variable with probability mass function (pmf) $(p_n)_{n \geq 0}$. The PGF of X , denoted by g (or sometimes g_X or h), is defined as:

This is a function $g : [0, 1] \rightarrow [0, \infty)$ given by

$$g(t) = \mathbb{E}[t^X] = \sum_{n=0}^{\infty} p_n t^n$$

This is a **Power Series**.

Properties

1. $g(1) = \sum_{n=0}^{\infty} p_n = 1$.
2. g is increasing on $[0, 1]$.
3. If X takes only finitely many values (e.g., $\text{Bin}(n, p)$), then g is a polynomial.
4. The radius of convergence of g is at least 1, i.e., $r \geq 1$ when r is the radius of convergence of g .

Differentiation and Probabilities

$g(t)$ is well-defined on the open disc $\{z : |z| < 1\}$. In particular, it is well-defined on $(-1, 1] \subseteq \mathbb{R}$. On the interval $(-1, 1)$, g is infinitely differentiable.

We can recover the probabilities from the derivatives at $t = 0$:

$$\begin{aligned} \left. \frac{d^n}{dt^n} g(t) \right|_{t=0} &= n! p_n \\ \implies p_n &= \frac{1}{n!} \left. \frac{d^n g}{dt^n} \right|_{t=0} \end{aligned}$$

Generating Functions of Discrete Random Variables

1. Constant Random Variable

Let $X \equiv c$ (constant).

$P(X=k)=1$

$$g_X(t) = t^c$$

2. Bernoulli(p)

Values: 0, 1 with probabilities $1 - p$ and p respectively ($0 < p < 1$).

$$g_X(t) = (1 - p)t^0 + pt^1 = 1 - p + pt = 1 - p(1 - t)$$

3. Binomial(n, p)

$$\begin{aligned} g_X(t) &= \sum_{k=0}^n \mathbb{P}(X = k)t^k = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} t^k \\ &= \sum_{k=0}^n \binom{n}{k} (pt)^k (1 - p)^{n-k} \end{aligned}$$

By the Binomial Theorem:

$$g_X(t) = (pt + (1 - p))^n = (1 - p(1 - t))^n$$

4. Geometric(p)

Let $X \sim \text{Geo}(p)$ with support $\{0, 1, 2, \dots\}$.

$$P(X = n) = (1 - p)^n p, \quad 0 < p < 1$$

The PGF is:

$$g_X(t) = \sum_{n=0}^{\infty} (1 - p)^n p t^n = p \sum_{n=0}^{\infty} (t(1 - p))^n$$

Using the geometric series sum $\sum r^n = \frac{1}{1-r}$:

$$g_X(t) = \frac{p}{1 - t(1 - p)} \quad \text{provided } |t(1 - p)| < 1 \implies t < \frac{1}{1 - p}$$

5. Poisson(λ)

Let $X \sim \text{Pois}(\lambda)$.

$$g_X(t) = \sum_{n=0}^{\infty} P[e^{-\lambda} \frac{\lambda^n}{n!}] t^n = e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda t)^n}{n!}$$

Using the Taylor expansion $e^x = \sum \frac{x^n}{n!}$:

$$g_X(t) = e^{-\lambda} \cdot e^{\lambda t} = e^{\lambda(t-1)}, t \in \mathbb{R}$$

PGF of Sums (I.I.D. Variables)

Recall $g_X(t) = \mathbb{E}[t^X]$.

$$g(t) = \sum_{n=0}^{\infty} p_n t^n = \mathbb{E}[t^X]$$

By Law of Unconscious Statistician

Alternative Derivation for Binomial

Let $X \sim \text{Bin}(n, p)$, where $X = \sum_{k=1}^n I_k$ and $I_k \sim \text{Bernoulli}(p)$ are i.i.d.

$$g_X(t) = \mathbb{E}[t^X] = \mathbb{E}[t^{\sum I_k}] = \mathbb{E}\left[\prod_{k=1}^n t^{I_k}\right]$$

By independence:

$$g_X(t) = \prod_{k=1}^n \mathbb{E}[t^{I_k}] = (\mathbb{E}[t^{I_1}])^n$$

Since $g_{I_1}(t) = 1 - p + pt$:

$$g_X(t) = (1 - p + pt)^n$$

6. Negative Binomial(k, p)

Let $X \sim \text{Neg-Bin}(k, p)$. X is the sum of k i.i.d. Geometric(p) variables: $X = \sum_{j=1}^k Y_j$.

$$g_X(t) = (g_Y(t))^k = \left(\frac{p}{1 - t(1 - p)}\right)^k \quad \text{for } |t| < \frac{1}{1 - p}$$

Identity we get

$$\sum_{n=0}^{\infty} P(X = n) t^n = \frac{p^k}{(1 - t(1 - p))^k}, \quad |t| < \frac{1}{1 - p}$$

($\iff$ using the series expansion for the Negative Binomial Distribution, where X is the number of failures before k successes):

$$\sum_{n=0}^{\infty} \binom{n + k - 1}{k - 1} (1 - p)^n p^k t^n = \frac{p^k}{(1 - t(1 - p))^k}$$

Sum of Bernoulli Variables (Binomial Distribution)

$$S_n = X_1 + X_2 + \cdots + X_n \quad (X_i \text{ is Bernoulli}(p))$$

$$S_n \sim \text{Bin}(n, p)$$

The probability generating function (PGF) is:

$$g_{S_n}(t) = (1 - p(1 - t))^n$$

Central Limit Theorem (CLT)

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0, 1)$$

Note: S_n must be integer valued or even non-negative.

Moment Generating Function (MGF) / Probability Generating Function (PGF)

$$g_X(t) = E[t^X] \quad \text{where } X \text{ is a non-negative integer valued random variable}$$

$$e^{sX} = t^X \quad \text{implies that } t = e^s$$

Definition of MGF: $M_X(t) = E[e^{tX}]$, $t \in \mathbb{R}$

Moment Generating Function (MGF)

The MGF is defined as:

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}$$

This is defined whenever the expectation [RHS] is finite.

Examples

1. **Bernoulli(p):**

$$M_X(t) = (1-p)e^{t \cdot 0} + pe^{t \cdot 1} = 1 - p + pe^t$$

2. **Binomial(n, p):** Using $X = \sum Y_k$ (i.i.d Bernoulli):

$$M_X(t) = (M_{Y_1}(t))^n = (1 - p + pe^t)^n$$

Properties of Moment Generating Functions

Note: If $X = \sum_{k=1}^n Y_k$, where $Y_1, Y_2, \dots, Y_n$ are i.i.d., then

$$\begin{aligned} M_X(t) &= (M_{Y_1}(t))^n \\ &= E[e^{tX}] \\ &= E\left[e^{t \sum_{i=1}^n Y_i}\right] \\ &= E\left[\prod_{i=1}^n e^{tY_i}\right] \quad \dots \end{aligned}$$

Normal Distribution and MGF

Standard Normal $X \sim N(0, 1)$

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - 2tx)} dx \end{aligned}$$

Completing the square in the exponent: $x^2 - 2tx = (x - t)^2 - t^2$.

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}((x-t)^2 - t^2)} dx \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2} dx \end{aligned}$$

The integral is the area under a $N(t, 1)$ PDF, which is 1.

$$M_X(t) = e^{t^2/2}, \quad t \in \mathbb{R}$$

Moments of $N(0, 1)$

Using the series expansion of $M_X(t)$:

$$e^{t^2/2} = \sum_{n=0}^{\infty} \frac{(t^2/2)^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{2^n n!} t^{2n}$$

The MGF Taylor series is $M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k$.

Comparing coefficients:

- Odd powers of t : The coefficients are 0, so $\mathbb{E}[X^{2m+1}] = 0$.
- Even powers t^{2m} :

$$\frac{\mathbb{E}[X^{2m}]}{(2m)!} = \frac{1}{2^m m!} \implies \mathbb{E}[X^{2m}] = \frac{(2m)!}{2^m m!}$$

Moment Generating Function and Moments

Series Expansion of $M_X(t)$

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= E\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right] \\ &\stackrel{\text{Interchange}}{=} \sum_{n=0}^{\infty} E\left[\frac{(tX)^n}{n!}\right] \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} E[X^n] \end{aligned}$$

Deriving Moments using MGF

Power Series Theory

$$\begin{aligned} \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0} &= \left. \frac{d^n}{dt^n} \left(\sum_{k=0}^{\infty} \frac{t^k}{k!} E[X^k] \right) \right|_{t=0} \\ &= \frac{E[X^n]}{n!} \cdot n! \\ &= E[X^n] \end{aligned}$$